



Monitoring of groundwater quality for drinking purposes using the WQI method and its health implications around inactive mines in Vemula-Vempalli region, Kadapa District, South India

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Received: 22 March 2020 / Accepted: 22 July 2020
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Abstract

A study was conducted to evaluate the groundwater quality and health risk assessment for drinking purposes around inactive mining areas in the rural regions of Vemula-Vempalli region using the water quality index (WQI) method. For this study, forty groundwater samples were collected from bore wells and analyzed for physical parameters and major cations and anions followed by standard methods of APHA. From the analytical results, most of the groundwater samples are in alkaline nature; EC, TDS, and TH values are below the permissible limit and major cations (Ca^{2+} , Mg^{2+} , K^+ , Na^+) & anions (Cl^- , HCO_3^- , SO_4^{2-} , F^- , NO_3^-) also within the permissible limit except fluoride and nitrate. Fluoride and nitrate in the groundwater range between 0.24–3 and 0.14–269 mg/L, respectively. Magmatic intrusions and water–rock interactions are main responsible for elevated fluoride in groundwater, and agricultural practices and usage of fertilizer are major responsible for higher nitrates in the groundwater. Piper diagram reveals that most of the samples belong to hydrochemical facies $\text{Ca}^{2+}\text{--Mg}^{2+}\text{--HCO}_3^-$ category in this region. A higher concentration of fluoride and nitrate in the groundwater may cause a serious impact on human health. Non-carcinogenic effects of F^- and NO_3^- were computed using total hazard index by adopting USEPA guidelines; THI values in drinking water range from 0.41 to 7.28 (adults), 0.41 to 7.38 (children) and 0.31 to 5.62 (infants); it reveals that children are more prone to the health impact than adults and infants. Overall assessment of WQI values (83.7–186.1 mg/L) shows that 7.5% (excellent), 80% (good), and 12.5% groundwater samples are very poor for drinking purpose in this region.

Keywords Health risk assessment · Fluoride · Nitrate · Inactive mines · Vemula-Vempalli · Kadapa District · A.P. South India

Introduction

Mining activity results in the creation of a lot of inactive/abandoned mine lands (AML). Groundwater quality and public health and safety concerns are the most common environmental issues from inactive mines in several countries with a long history of mining like India (Mhlongo et al. 2013). Due to the groundwater contamination, toxic ions are of a huge health problem noticed in inactive/abandoned mines, especially in India. Toxic elements like fluoride & nitrate are released into groundwater and can be accumulation in crops and may cause significant effect on human

health (Davies 1983). Of this, the most common environmental issue related to the inactive/abandoned mines in several countries mainly human health (Sphiwe and Francis 2015). Quality of groundwater is of major concern in deciding the applicability of water quality evolution for drinking purposes, and it must be devoid of harmful elements, living & non-living organisms in groundwater, excessive mineral dissolution which may cause an impact on human health (Muralidhara Reddy et al. 2013). Geogenic and anthropogenic sources from agriculture practices and dissolution of rock and mine water pollution contaminated soil mainly affects on groundwater quality (Guo et al. 2007; Brindha and Elango 2013; Adimalla et al. 2018; Adimalla and Venkatayogi 2018; Li 2016; Wu et al. 2017; Karunaidhi et al. 2019). Freshwater has become a scanty commodity due to overuse, and water is getting contaminated. Groundwater is widely used for drinking, agriculture, and industrial purposes (Adimalla and Wu 2019; Wu and Sun 2016). Over around world

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